

Assignment No. 01
Assigned Date : 2-05-2024
Due Date : 09-05-2024 11:00PM

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Attempt All Questions

1. Following are programs expressed as English statement. Write only the programing statements for actual program and their output.
The variable x starts with the value 10
The variable y starts with the value 2
The variable z starts with the value 4
Store the value of x times y in x
Store the value of y times z in y
Add x and z and store result in y
What is the value of y if output on screen
2. Following algorithm has an error. The program is supposed to ask the user for the length and width of a rectangular room, and then display the room's area. The program must multiply the width by length in order to determine the area.
area = width into length
Display "What is Room's Width"
Input width
Display "What is Room's Length"
Input length
Display area
3. Which of the following is a Character Literal
'A'
"A"
4. Assuming the size of data types required in memory is 1B,4B,8B for Character, integer and double respectively. How many bytes the following declaration required

int x,y,z;
char a,b;
double j,k;
5. How would you consolidate the following definitions into one statement.
int x = 7;
int y = 20;
int z = 28;

6. How many operands does each of the following types of operator require?

Unary , Binary and Ternary

7. How the group of statements are enclosed to treat as a block.

8. Assume $w = 5$, $x = 4$, $y = 8$, and $z = 2$. What value will be stored in result in each of the following statements?

result = $x + y$;

result = $z * 2$;

result = y / x ;

result = $y - z$;

result = $w \% 2$;

9. Write expressions for the following algebraic expressions.

$$y = 6x$$

$$a = 2b + 4c$$

$$y = x^2$$

$$g = \frac{x+2}{z^2}$$

$$y = \frac{x^2}{z^2}$$

10. Design an algorithm to find the perimeter and area of rectangle. Formula used to find the solution are

Perimeter = $2 * (\text{length} + \text{width})$

Area = $\text{length} * \text{width}$

11. Design an algorithm to calculate the sales tax and price of an item sold in a particular state. The sales tax is calculated as follows. The state's portion of sales tax is 4% and the city's portion of sales tax is 1.5%. if cost of item is more than 50000 then a luxury tax of 10% is applicable.

12. Given the radius, in inches, and price of pizza, design an algorithm to find the price of pizza per square inch.

13. To get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a complete program that stores the following values in five different variables: 28 , 32 , 37 , 24 and 33. The program should first calculate the sum of these five number and then stores then in a different variable named sum. Then the program should display the average on screen. [Use data types correct]

